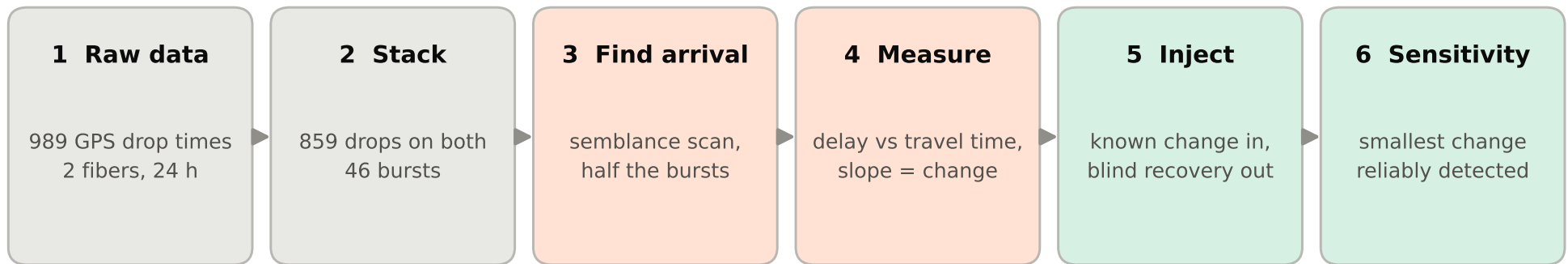


From raw fiber data to a sensitivity number

Two fibers in one borehole, one repeated source, 24 hours



Nano cemented, shallow

927 m of fiber · 30–60 Hz · ~2950 m/s

Deep wireline, reversing

6532 m of fiber, two legs · 15–30 Hz · ~1547 m/s

The cemented fiber resolves a 1% velocity change; the wireline outbound branch resolves 0.5%

The wireline arrival travels 11.8× longer, but its timing is 7.6× noisier — so the gain is 2×, not 12×

Raw data to burst stacks

In: raw fiber recordings, plus 989 GPS times of the weight drops.

For each drop a 3.5 s window is cut from every channel — 0.5 s before, 3.0 s after. A drop is kept only if that window fits inside its file.

Out: one averaged waveform per burst, per channel.

Single drops vary by $\sim 20\times$ in signal power, so they are averaged in groups. Drops >60 s apart are separate bursts.

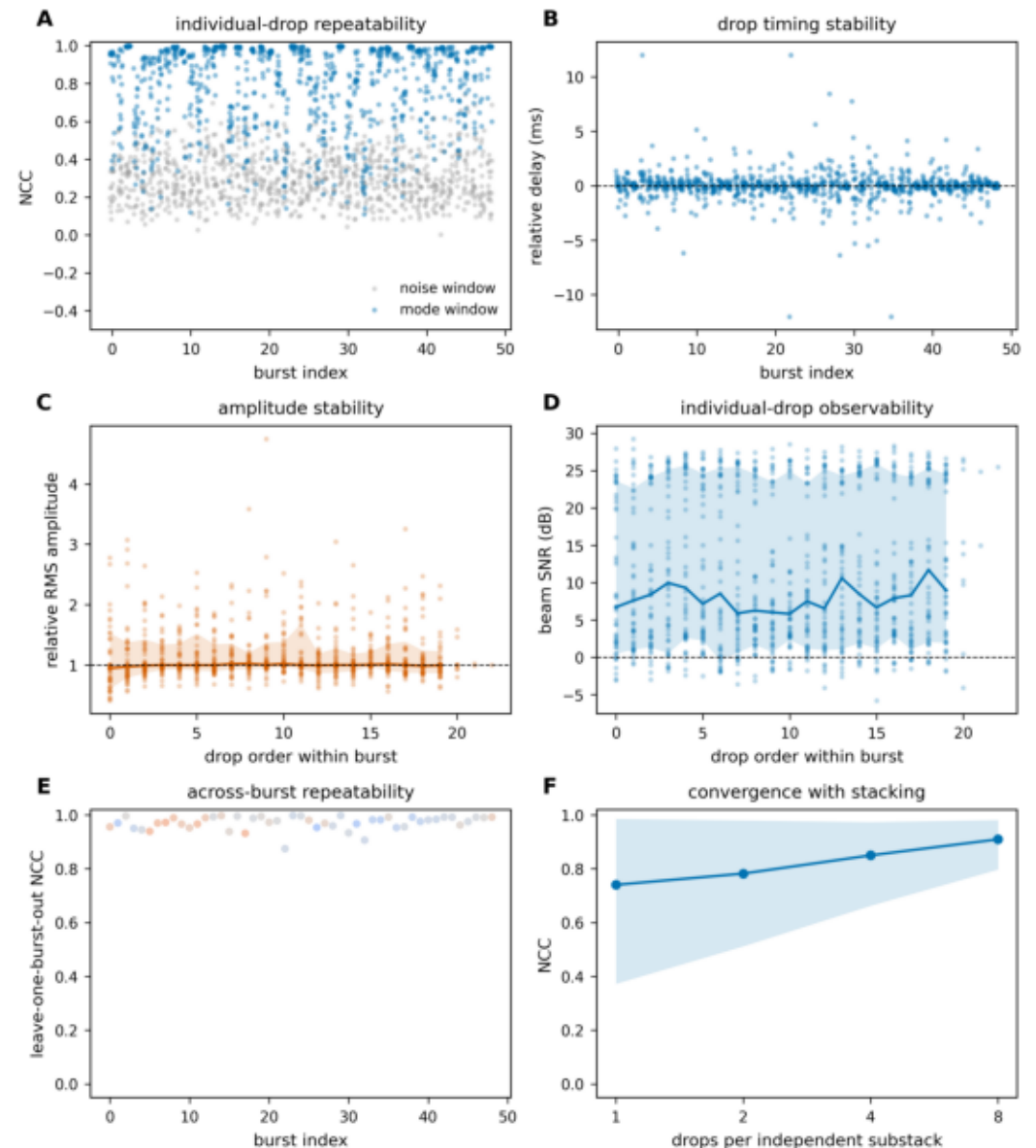
The Deep fiber stopped after burst 45, so the last three bursts are Nano-only. That is why 49 bursts becomes 46 — not any quality cut.

46 bursts analysed

859 drops, both fibers

23.96 h survey duration

30 min burst cadence



Finding the arrival

An arrival moving along the fiber is a straight line in distance vs time.

Describing it takes two numbers: a start time and a speed.

Shift every channel by its predicted arrival time and sum. A correct guess adds up; a wrong one cancels. The peak of that ratio picks the trajectory.

The search uses half the bursts only, so sensitivity is not measured on the same data that chose what to look at.

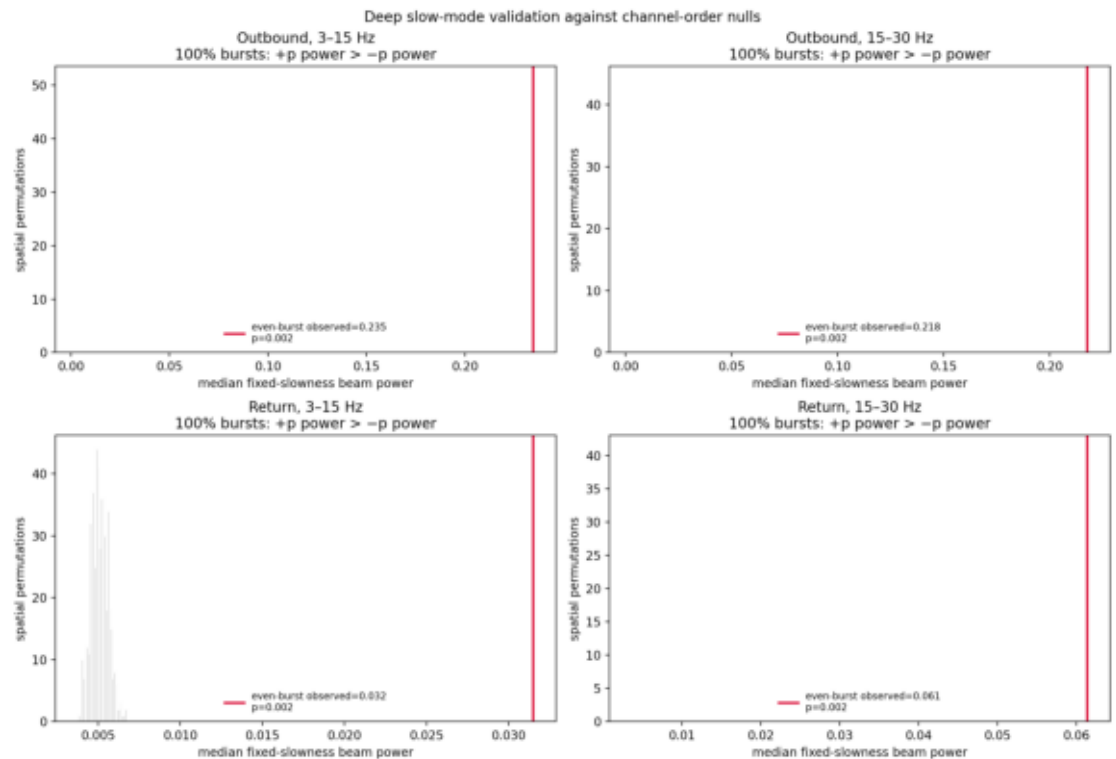
Scrambling channel order 499 times never reproduces the real arrival.

2975 m/s Nano

1544.6 m/s Deep outbound

1549.7 m/s Deep return

0.3% legs agree to



Measuring a velocity change

Compare one burst against the average of all the others.

Measure the delay at several positions along the fiber — per channel for Nano, per 400 m aperture beam for Deep, which needs the extra stacking at 3 km.

If the source fired early, every delay shifts by the same amount.

If the medium sped up, the delay grows with distance travelled.

So fit a line through delay vs travel time. The intercept absorbs source timing; the slope is the velocity change.

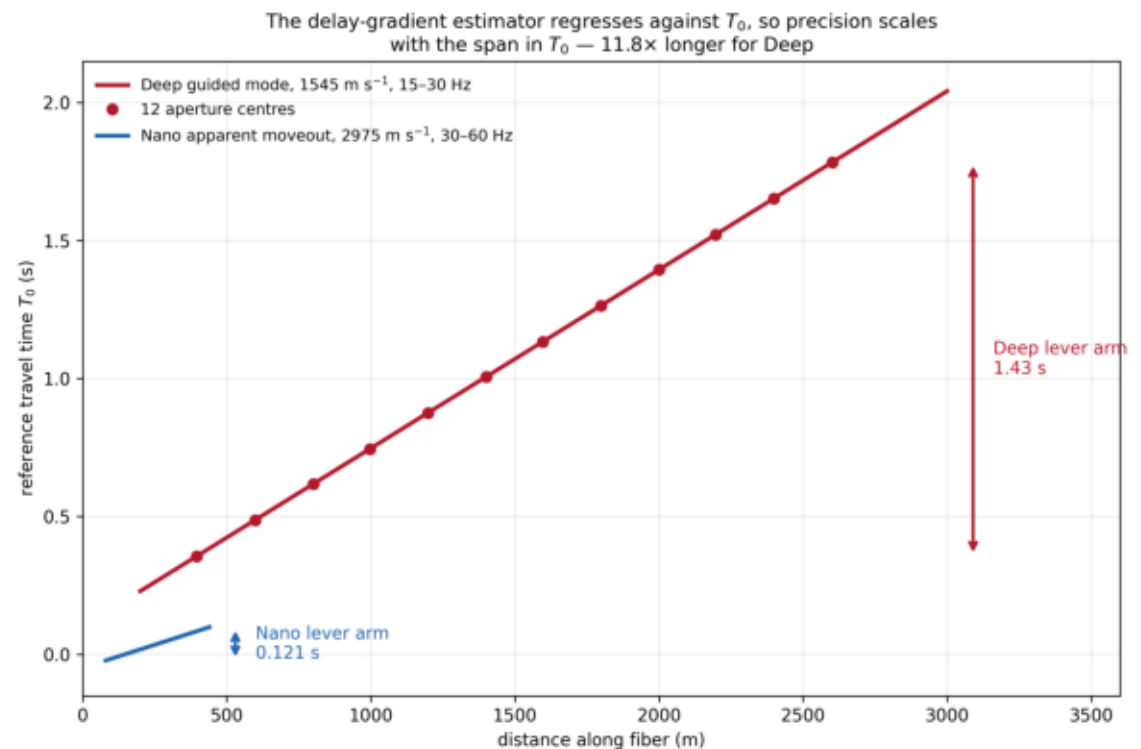
Precision scales with the span in travel

time — the lever arm.

0.121 s Nano lever arm

1.428 s Deep lever arm

11.8x ratio



How small a change is detectable

**Do not estimate this from theory —
inject known changes into real data.**

Each channel is shifted exactly as a real velocity change would shift it, at 15 levels from $\pm 0.01\%$ to $\pm 1\%$, plus zero.

The recovery code never sees the injected value. A sealed answer key is joined only at the end.

The noise floor is what comes back when nothing was injected. The reliable level is the smallest injection recovered with the right sign, above that floor, 95% of the time.

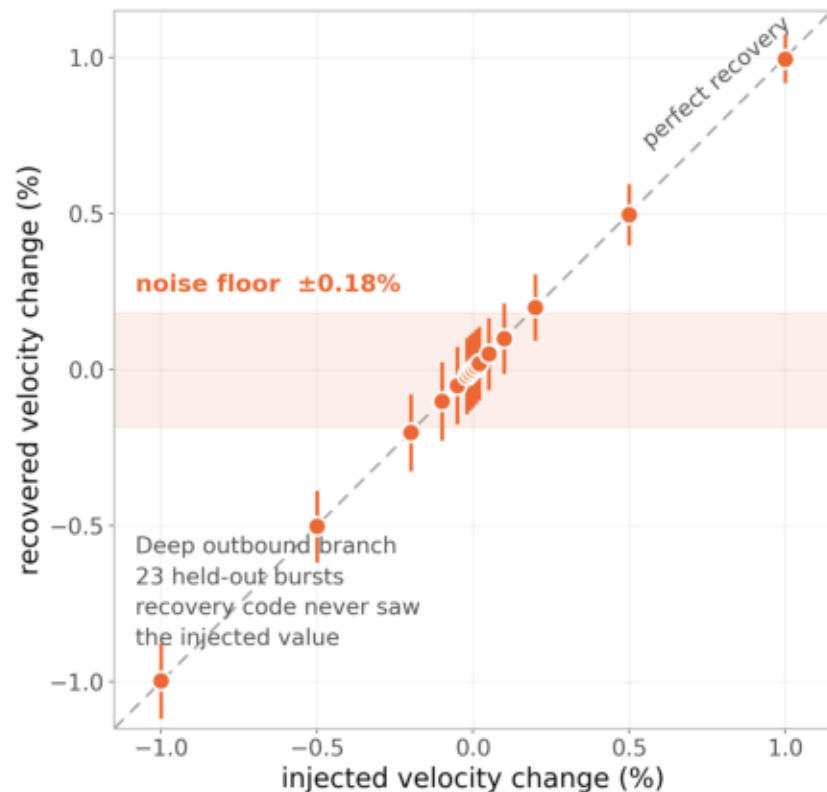
15 injection levels

345 trials per leg

23 held-out bursts

0.18% outbound floor

Known changes injected into real data are recovered blind down to a few tenths of a percent



RESULT

What each installation resolves

Cemented fiber: 1%.

Wireline outbound branch: 0.5%.

Wireline return branch: 1%.

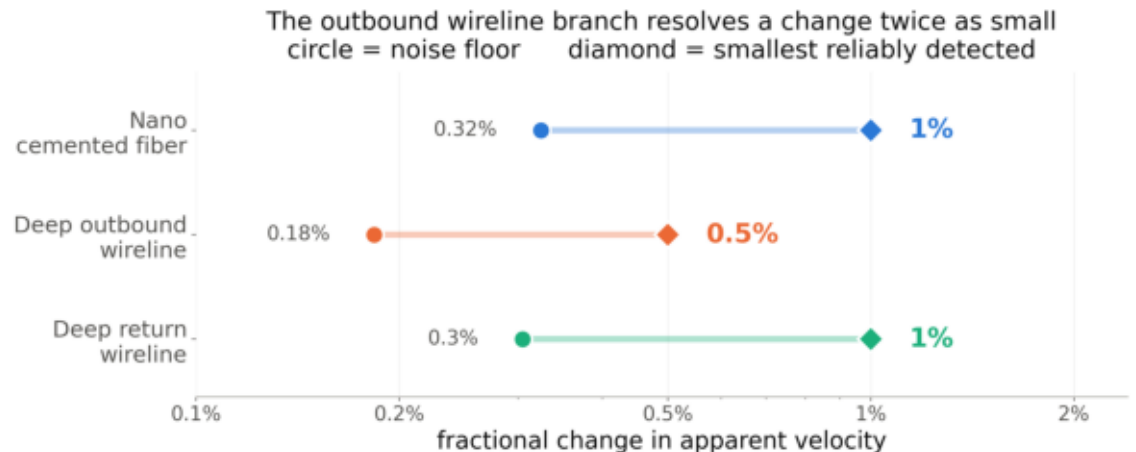
The improvement therefore belongs to the outbound branch, not to the wireline installation as a whole.

Why only 2×, when the lever arm is 11.8× longer?

Because the deep arrival's timing is 7.6× noisier burst to burst. $11.8 \div 7.6 = 1.55$, exactly the observed ratio of scatter.

Geometry helps only in proportion to timing repeatability. That is the result that transfers to other experiments.

1.0%	Nano
0.5%	Deep outbound
1.0%	Deep return
7.6×	timing penalty



Why we believe it

Synthetic test. A noiseless wave

perturbed two independent ways recovers the same answer to five significant figures — the injection is faithful.

Timing control. Shift everything by

5 ms: the intercept absorbs 4.87 ms and the velocity estimate moves 7×10^{-7} . Source timing does not leak in.

Permutation test. Scramble channel

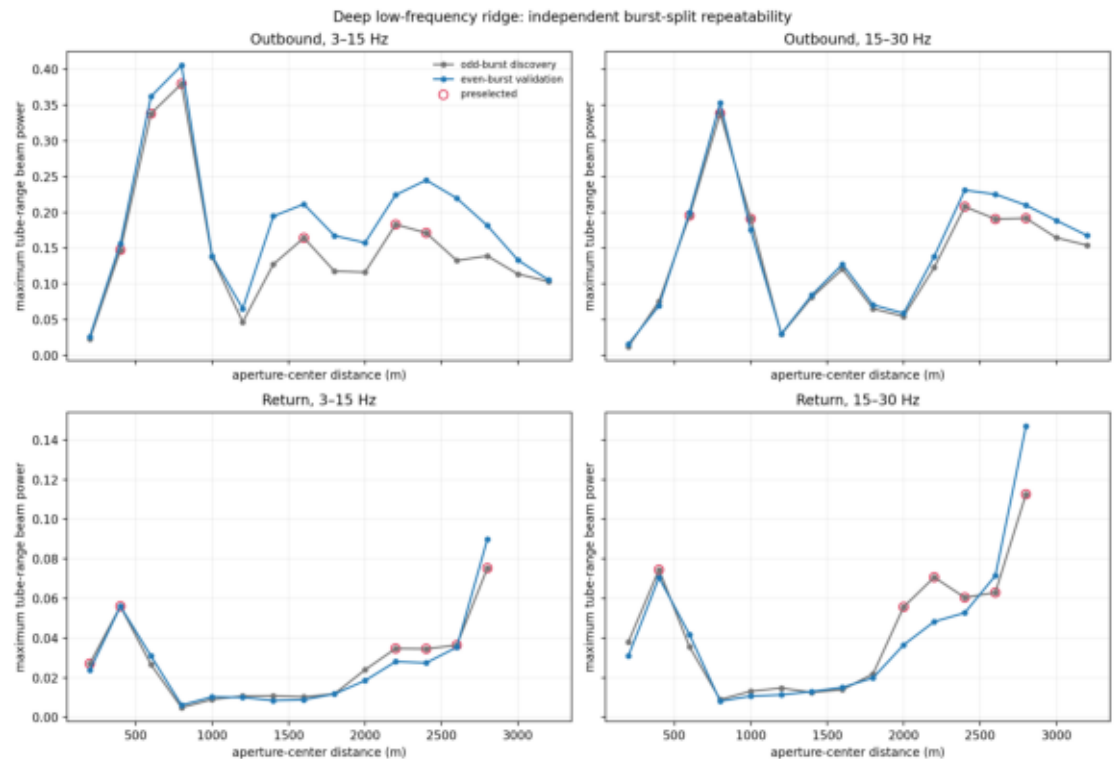
order 499 times: zero scrambles reach the observed beam power, in all four tests.

Jackknife. Drop each aperture, then

each burst. Nothing moves the result by as much as the noise floor.

One control failed, and is reported.

The perturbed-trajectory test recovers injections even pointed at the wrong arrival, because the injection shifts whole traces. It fires the same on both legs, so it cannot discriminate — the return branch is left unclassified rather than explained after the fact.



What this does not show

Not a formation velocity. The measured

quantity is the apparent speed of a guided mode along the fiber. Both observables are dispersive; neither is a clean body wave.

No depth resolution. Position is

distance along fiber. The depth mapping is provisional and no result depends on it.

No tidal detection. Fitting a solid-

Earth-tide model to the per-burst velocities gives a clean null, with a 95% upper limit $\sim 18\times$ above the tidal signal expected at this site.

The limit is the 24-hour survey length, not the instrument: over one cycle a daily signal is hard to separate from drift.

